

Interfacing Macaque Cortex with Language Models for Generative Neural Readout

Victoria Bosch^{*1} Daniel Anthes^{*1} Adrien Doerig² Peter König¹ Tim C Kietzmann¹

¹ Institute of Cognitive Science, Osnabrück University

² Department of Psychology and Education, Freie Universität Berlin

^{*} Shared first authorship

victoria.bosch@uni-osnabrueck.de, tim.kietzmann@uni-osnabrueck.de

Abstract

Recent advances in neural decoding have moved beyond categorical readouts toward richer, semantically grounded representations, for example by mapping brain activity elicited by natural scenes to language embeddings of image captions. While effective, such approaches remain limited to static representations. To enable more flexible and interactive decoding of neural data, recent work has proposed interfacing brain activity directly with large language models (LLMs), allowing neural signals to condition open-ended language generation. However, these approaches have so far been restricted to human fMRI and time-averaged signals. Here, we extend this approach to intracranial recordings from macaque visual cortex during passive viewing in the THINGS dataset (Papale et al., 2025). We show that time-resolved neural activity is decodable into natural language, enabling generative readout of visual content. These preliminary results demonstrate that LLM-based neural decoding interfaces generalize beyond human fMRI and suggest the presence of structure in non-human primate visual representations that can be aligned with natural language embeddings.

Neural decoding from macaque visual cortex with CorText

We adapt the CorText framework (Bosch et al., 2025) to macaque electrophysiology by embedding neural activity as a sequence of spatial and temporal brain tokens, which are injected into a pretrained LLM to condition language generation (see Fig. 1). This enables decoding of neural responses via natural language queries (Fig. 1B). To quantify decoding accuracy, we use the cosine similarity between QWEN-3 embeddings of predicted captions and ground-truth captions (Fig. 1C). These results show that in both monkeys, caption decoding performance is significantly better than that of the control model (Control $\mu_{cap} = 0.45$; Monkey F, $\mu_{cap} = 0.54$ vs control model: $p < 0.001$; Monkey N, $\mu_{cap} = 0.54$; $p < 0.001$; independent permutation tests, $n=10000$). To perform annotation-free evaluation, we also measure alignment of the predicted caption with the stimulus image, we use CLIPScore (Hessel et al., 2021) (Fig. 1D) (Control $\mu_{cap} = 0.16$; Monkey F, $\mu_{cap} = 0.20$ vs control

model: $p < 0.001$; Monkey N, $\mu_{cap} = 0.19$; $p < 0.001$; independent permutation tests, $n=10000$). ‘Ceiling’ values indicate the average CLIPScore of the ground-truth caption with the stimulus image.

Discussion

We demonstrate that CorText-style neural interfaces extend to macaque electrophysiology and support generative decoding of visual content from V1, V4, and IT.

Prior work has demonstrated that macaque visual cortex supports accurate decoding of object identity and aligns with deep visual feature spaces, including zero-shot generalization (Conwell et al., 2025; O’Connell et al., 2019) and image reconstruction (Dado et al., 2024; Fei et al., 2025). These approaches have provided important insights into the structure of neural representations, but remain limited to fixed output spaces or image reconstructions. In contrast, our approach interfaces neural activity directly with a LLM, enabling open-ended and queryable semantic readout. This allows neural responses to be interrogated flexibly through natural language, rather than being constrained to predefined labels or feature spaces.

One limitation of this preliminary work is the restricted task range of the current VQA dataset, which is based primarily on model-generated captions. Future work should extend the THINGS annotations to include more targeted queries, such as numerosity, spatial relations, and fine-grained attributes, enabling more precise assessment of decodable information. This also points to a potential improvement in evaluation metrics: the decoding scores are significant but modest, likely due to a high sensitivity of QwenScore to nouns. While features such as colour and material are decoded well, mistaken object descriptions may result in a bad score. More precise type evaluation would alleviate this limitation. In addition, further analysis of the neural-language interface could clarify the contributions of different regions and timepoints, for example, through attention or gradient-based attribution methods. Finally, causal perturbations of neural activity or neural tokens may provide a route toward identifying which components of the signal drive specific aspects of the generated outputs. Together, these results highlight



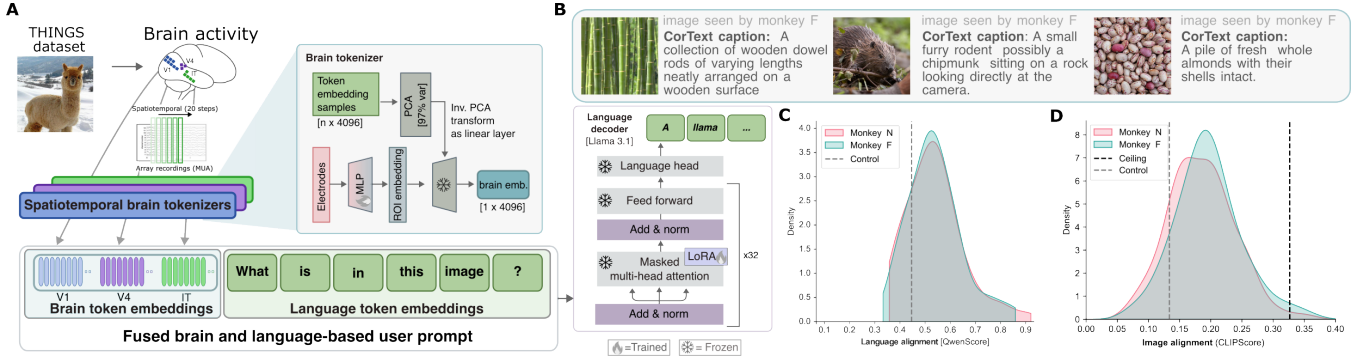


Figure 1: **A.** CorText model architecture applied to macaque array recordings. **B.** Example captions of the test set for CorText trained on monkey F. **C.** Alignment between predicted and ground-truth image captions of the test set. **D.** Alignment between predicted captions and stimulus images of the test set.

the potential of language-interfaced neural decoding as an interactive diagnostic tool, enabling targeted and flexible probing of neural activity across imaging modalities and species.

Methods

Neural data We analyze data from the “Things Ventral Stream Spiking Dataset” (TVSD) consisting of multi-unit activity (MUA) recorded from two macaques during passive viewing of the THINGS dataset (Papale et al., 2025). Neural data was recorded simultaneously from 16 Utah arrays (1024 channels) across V1, V4, IT. The dataset comprises object-centric natural images spanning a wide range of semantic categories, with predefined train/test splits that we adopt throughout. The original test set consists of 30 repeats of 100 stimuli, whereas we average the repeats. The data is z-scored per timestep across the training set. The neural responses are binned at 10 ms resolution in a stimulus-locked window from 0-200 ms post-stimulus onset. This yields a sequence of time-resolved neural responses for each array, which we will treat as “brain tokens” (320 per trial).

Visual Question-Answering Dataset For each image shown in THINGS (Hebart et al., 2023), we generate five independent captions using the LLaVA-34B vision-language model (Liu et al., 2023). Captions are produced using a fixed prompt: “Give a caption for this image. It should be brief, no longer than 1-2 sentences and give an accurate description of the image. Include no other information or background information.” The prompt is selected by qualitatively verifying of a set of random generated captions, and yielding a set of diverse, naturalistic descriptions per image, which we use for training and evaluation. Using multiple captions per image reduces bias toward a single phrasing and better captures semantic variability. To construct a visual question-answering dataset defined over the THINGS stimuli, we combine these captions with a set of caption questions, such as “Give a concise and descriptive caption of this image.”

Model Architecture CorText is an LLM-based brain-language fusion architecture. It uses Llama 3.1 8B-Instruct as backbone (Grattafiori et al., 2024) and a set of brain tokenizers that map neural data into the language decoder (see Fig. 1A; see section Brain tokenizers). Llama 3 is an instruction-finetuned generative language model, which has been finetuned using supervision and reinforcement learning from human feedback (RLHF). We keep the model weights quantized in 4-bit precision for memory efficiency.

Brain Tokenizers To enable multimodal fusion between neural data and language, we use one MLP with two hidden layers (dimensionality 32) per array and time bin (see Data). Each of these MLPs will learn a mapping from the neural data into a 4096-dimensional vector, i.e. “a brain embedding”. To reduce the encoder size, we use a frozen low-rank linear projection layer to embed neural data of each brain token, according to the approach in (Bosch et al., 2025). After embedding the neural data for a trial using brain tokenizers, the brain embeddings are prepended to a Llama tokenizer-embedded question. The sequence is formatted into Llama’s instruction prompt structure with instruction tokens, yielding a multi-modal input sequence of brain and text embeddings.

Training Model training proceeds in two phases. During pretraining (5 epochs), only the brain tokenizers and layer normalization are trained end-to-end with the frozen language decoder. During finetuning (3 epochs), we apply QLoRA to the decoder’s query and value projections (rank 16, $\alpha = 16$, dropout 5e-4), while keeping the base model frozen. Optimization uses AdamW-8bit with batch size 64 and learning rates of 2e-3 (pretraining) and 2e-5 (finetuning). We apply dropout and L2 regularization to the brain tokenizer weights (0.2 / 0.15, respectively). In both phases, the model is trained with a cross-entropy loss on generated answers conditioned on neural context embeddings. As a control, we train a model on shuffled neural data, effectively removing neural signal and yielding a language-only baseline.

Acknowledgements

The authors acknowledge support by the ERC stg grant 101039524 TIME (Bosch, Doerig, Kietzmann), SNF grant n.203018 (Doerig), RTG GRK2340 DFG (Anthes). Compute resources for this project are in part funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation), project number 456666331. ChatGPT 5.4 was used for proofreading and light editing of the text in this manuscript.

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